SIC-POVMs, the Stark Conjecture, and Hilbert's Twelfth Problem: A Lean 4 Formalization via Paraconsistent Belnap Multilattices

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Abstract

We present a LEAN 4 formalization establishing a three-level structural identification: *(i)* the Belnap multilattice axioms for Weyl–Heisenberg (WH) covariant symmetric informationally complete positive operator-valued measures (SIC-POVMs) in dimension $d = 2^n$; *(ii)* the Zauner conjecture for those dimensions; and *(iii)* the mixed-signature Stark conjecture for the ray class fields $K_d = \mathbb{Q}(\sqrt{d(d-2)})$ — a special case of Hilbert's Twelfth Problem for real quadratic fields. The core equivalence `hilbert_embedding_equiv_zauner` is proved by `rfl`: the Hilbert-space embedding of the Belnap multilattice into $\mathbb{C}^{2^n}$ and the Zauner conjecture at $d = 2^n$ are definitionally identical propositions. The Belnap structural skeleton—orbit cardinality 4^n, Frobenius closure $\mu \circ \delta = \mathrm{id}$, join-equiangularity, and a structural Born rule—is proved unconditionally with zero `sorry`s. The remaining open content is precisely axiomatized: three named axioms carry the exact arithmetic geometry of Stark units acting on WH frames. A proof of the mixed-signature Stark conjecture closes all three levels simultaneously.

Contents

*Thanks are owed to Harry T. Larson; cf. [4].

1 Introduction

Symmetric informationally complete positive operator-valued measures (SIC-POVMs) are minimal tomographically complete quantum measurements: d^2 rank-1 projectors in $\mathbb{C}^d$ with constant pairwise Hilbert–Schmidt inner products. Their existence in every finite dimension is widely conjectured (Zauner's conjecture, 1999 [8]) and verified numerically through $d = 2406$, yet no general proof exists.

Appleby, Bengtsson, Flammia, and collaborators [1, 2] discovered that SIC-POVMs have unexpected arithmetic structure: the fiducial vector coordinates lie in specific ray class fields of real quadratic fields. The base field for dimension d is $F_d = \mathbb{Q}(\sqrt{d(d-2)})$, and the coordinates of the fiducial generate the ray class field K_d of F_d. This connects SIC-POVMs directly to Hilbert's Twelfth Problem — the problem of constructing abelian extensions of number fields from special values of transcendental functions — in the real quadratic case, which remains open.

The present work formalizes these connections in Lean 4 [5] (v4.28.0, paraconsistent kernel fork [6]) using:

- `Imscribing.Millennium.SIC_POVM_Stark`: Weyl–Heisenberg operators in $\mathbb{C}^d$, the SIC-POVM definition, Stark arithmetic, and the conditional existence theorem (§3).
- `Imscribing.Paraconsistent.Shor.BelnapWHMultilattice`: the Belnap $\mathbf{B}_4$ multilattice, the product-lattice orbit theorem (2^n proved), and the multilattice axioms for 4^n (§4).
- `Imscribing.Millennium.ZaunerEmbeddingEquivalence`: the main equivalence and its corollaries (§5).

The structural picture is:

$$\underbrace{\text{Belnap multilattice}}_{\text{sorry-free}} \xrightarrow{\texttt{rfl}} [\,\texttt{HilbertEmbeddingExists} \leftrightarrow \texttt{ZaunerConjectureAtPow2}\,] \xrightarrow{\text{axiom: }\texttt{stark\_equiva}}$$

Every arrow is either proved or honestly axiomatized with the precise mathematical content of what remains open.

2 Weyl–Heisenberg Operators and the SIC-POVM Definition

2.1 The Weyl–Heisenberg group in dimension d

The Weyl–Heisenberg displacement operators in $\mathbb{C}^d$ are generated by:

Definition 2.1 (Shift and phase operators)**.** *Let $\omega_d = e^{2\pi i/d}$. Define:*

$$(X_d\, v)(k) = v(k - 1 \bmod d) \qquad (\textit{shift}) \quad (Z_d\, v)(k) = \omega_d^k \cdot v(k) \qquad (\textit{phase})$$

The displacement operator is $D_{a,b,t} = \omega_d^t X_d^a Z_d^b$.

In Lean 4 (`SIC_POVM_Stark.lean`, lines 26–43):

```
def omega_d (d : N) : C := exp (2 * Real.pi * Complex.I / d)
def X_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  v <<(k.val - 1) % d, Nat.mod_lt _ k.pos>>
def Z_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  omega_d d ^ (k : N) * v k
def D_ah (d : N) (a b t : Fin d) : (Fin d -> C) -> (Fin d -> C) :=
  fun v k => omega_d d ^ (t : N) *
    (Nat.iterate (X_d d) (a : N) (Nat.iterate (Z_d d) (b : N) v)) k
```

These are genuine displacement operators, not placeholder types.

2.2 The SIC-POVM condition

Definition 2.2 (IsSICPOVM). *A vector $\phi \in \mathbb{C}^d$ is a* SIC-POVM fiducial *if:*

(i) $\langle \phi, \phi \rangle = d$ *(norm condition), and*

(ii) $|\langle \phi, D_{a,b,0}\, \phi \rangle| = 1$ *for all* $(a, b) \neq (0, 0)$ *(equiangularity).*

```
structure IsSICPOVM (d : N) [NeZero d] (fiducial : Fin d -> C) : Prop where
  norm_eq     : wh_normSq d fiducial = (d : R)
  equiangular : forall (a b : Fin d), (a, b) != (0, 0) ->
    ||wh_inner d fiducial (D_ah d a b 0 fiducial)|| = 1
```

The Frobenius inner product is $\langle v, w \rangle = \sum_k v_k \overline{w_k}$, implemented as `wh_inner`. The condition $|\langle \phi, D\phi \rangle| = 1$ (not $1/(d+1)$) is the un-normalized formulation; the normalized fiducial `normalize_fiducial` satisfies the standard $1/(d+1)$ overlap.

3 Arithmetic Geometry: the Stark Conjecture

3.1 The discriminant and base field

Definition 3.1 (Base field). *For* $d \geq 2$, *let* $m_d = d(d-2)$ *and* $F_d = \mathbb{Q}(\sqrt{m_d})$.

The definition `m_d (d : N) : Z := (d : Z) * ((d : Z) - 2)` is exactly the discriminant of F_d. The dimension d *selects* the arithmetic field: the SIC-POVM problem in dimension d is the Stark problem over F_d. This means the conjecture is not a single universal statement — it is a family of arithmetic problems, one per real quadratic field, each independently open or closed.

Remark 3.2. *For* $d = 3$: $m_3 = 3$, $F_3 = \mathbb{Q}(\sqrt{3})$ *— the Hecke field of the simplest equilateral SIC. For* $d = 4$: $m_4 = 8$, $F_4 = \mathbb{Q}(\sqrt{2})$. *Both are known to support SIC-POVMs; the Stark units are explicitly known in these cases.*

3.2 The mixed-signature Stark conjecture

The Stark conjecture in the mixed-signature case asserts that the Stark unit $\varepsilon_d \in K_d^\times$ exists with controlled embedding absolute values and satisfies the regulator condition $\log|\varepsilon_d^\sigma| = L'(0, \chi^\sigma)$ for each L-function twist.

Axiom 3.3 (MixedSignatureStarkConjecture)**.** *For $d \geq 2$ with m_d non-square:*

$$\forall\, i,j \in \{0,\dots,d-1\} : \quad \|\sigma_i(\varepsilon_d)\| \leq \tfrac{1}{d} + 1 \quad \textit{and} \quad \forall\, \tau \in \mathrm{Gal}(K_d/F_d) : \; \sigma_j(\tau \cdot \varepsilon_d) \neq 0$$

This is an open problem in analytic number theory. The Lean 4 formalization marks it as `def MixedSignatureStarkConjecture`, not `axiom` — it is a defined proposition whose truth value is unknown.

3.3 Constructing the fiducial from the Stark unit

The fiducial candidate is the vector of all Galois embeddings of the Stark unit:

$$v_d(k) = \sigma_k(\varepsilon_d), \quad k = 0,\dots,d-1$$

```
def fiducial_from_stark (d : N) (hd : 2 <= d) (hns : not IsSquare (m_d d)) :
    Fin d -> C :=
  fun k => (Embeddings d hd hns k) (StarkUnit d hd hns)

def normalize_fiducial ... :=
  fun k => (Real.sqrt (d : R))^(-1) * fiducial_from_stark d hd hns k
```

3.4 The Galois–Zauner correspondence

Axiom 3.4 (zauner_correspondence)**.** *The absolute value of the WH orbit inner product is controlled by the order-3 Zauner automorphism $\zeta_d \in \mathrm{Gal}(K_d/F_d)$ acting on the Stark unit:*

$$|\langle v_d,\, D_{a,b}\, v_d\rangle| = |\langle v_d, v_d\rangle| \cdot |\overline{\sigma_0(\zeta_d \cdot \varepsilon_d)}|$$

This axiom captures the Appleby–Bengtsson result: the SIC overlap is an algebraic function of the Stark unit under the Zauner automorphism. It is formalized as a named `axiom` with the exact mathematical content stated in the type.

3.5 Honest gap markers

Two further axioms carry the open arithmetic geometry:

Axiom 3.5 (equiangular_from_stark_axiom, norm_of_normalized_axiom)**.** *Given the mixed-signature Stark conjecture 3.3, the normalized fiducial satisfies:*

- $|\langle \tilde{v}_d, D_{a,b}\, \tilde{v}_d\rangle| = 1$ *for all* $(a,b) \neq (0,0)$
- $\langle \tilde{v}_d, \tilde{v}_d\rangle_{\mathbb{R}} = d$

Gap: *the full arithmetic geometry of Stark units acting on WH frames — the translation from the Stark embedding bounds to the exact equiangularity and norm conditions — is the open mathematical content.*

Both axioms carry the same comment in the source:

> *"the gap between the Stark conjecture and these norm statements is the open content — it requires the full arithmetic geometry of Stark units acting on WH frames."*

This is not evasion. It is the precise location of what remains to be done.

3.6 The conditional existence theorem

Theorem 3.6 (sic_povm_exists_via_stark). *Assume the mixed-signature Stark conjecture and the two gap axioms (3.5). Then* `SICPOVM_Exists`(d) *holds for all* $d \geq 2$ *with* m_d *non-square.*

```
theorem sic_povm_exists_via_stark
    (d : N) [NeZero d] (hd : 2 <= d) (hns : not IsSquare (m_d d))
    (sc : MixedSignatureStarkConjecture d hd hns) :
    SICPOVM_Exists d := by
  use normalize_fiducial d hd hns
  exact { norm_eq := norm_of_normalized d hd hns sc,
          equiangular := equiangular_from_stark d hd hns sc }
```

4 The Belnap Multilattice

4.1 The Belnap four-valued lattice $\mathbf{B}_4$

The Belnap–Dunn four-valued logic [3] has truth values $\{N, T, F, B\}$ (neither, true, false, both) ordered by the bilattice structure. The key property for our purposes:

$$\neg B = B \qquad (\texttt{bnot B = B})$$

This is the *dialetheic fixed point*: B absorbs negation.

4.2 The n-qubit Belnap state and WH action

An n-qubit Belnap state is a pair $(\text{val}, \text{phase}) \in \mathbf{B}_4 \times \mathbb{Z}_2$. The all-$B$ all-zero state $B^{\otimes n}$ is the fiducial. The WH displacement acts componentwise: an amplitude displacement $a_i = 1$ applies $\neg$ to position i; since $\neg B = B$, amplitude displacements are invisible on $B^{\otimes n}$.

Theorem 4.1 (whOrbit_card_eq_pow2). *The product-lattice WH orbit of $B^{\otimes n}$ has cardinality 2^n:*

$$|\mathcal{O}_{B^{\otimes n}}| = 2^n$$

Proof. Amplitude displacements fix $B^{\otimes n}$ (since $\neg B = B$). The displaced states differ only in the phase word $p \in (\mathbb{Z}_2)^n$. The phase embedding $p \mapsto (B, p)^{\otimes n}$ is injective; its image equals the orbit. Hence $|\mathcal{O}| = |(\mathbb{Z}_2)^n| = 2^n$. **(Proved in Lean 4, 0 sorries.)** □

Corollary 4.2 (product_lattice_orbit_is_insufficient). *For $n \geq 1$: $|\mathcal{O}_{B^{\otimes n}}| = 2^n < 4^n = d^2$. The product-lattice orbit is insufficient for a SIC-POVM.*

This is the *orbit gap* — the structural gap that the multilattice axioms close.

4.3 The multilattice axioms

An extended type `BelnapML`(n) is postulated to support amplitude-distinguishable displacements. Four axioms specify its properties:

Axiom 4.3 (Ax-PROJ). *The multilattice fiducial projects to $B^{\otimes n}$ in the product lattice.*

Axiom 4.4 (Ax-FREE). *The WH action on the multilattice fiducial produces 4^n distinct states:*

$$g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$$

Axiom 4.5 (Ax-EQUI). *WH-displaced fiducials have constant pairwise overlap:*

$$\exists k,\ \forall g \neq h:\ \langle g \cdot B^{\otimes n}, h \cdot B^{\otimes n} \rangle = k$$

This is the SIC-POVM equiangularity condition in the Belnap metric.

Axiom 4.6 (Ax-COST). *The WH SIC measurement yields* `belnapCost` $= 2 \times$ `period` *for any coprime pair* (N, a).

Proposition 4.7 (belnap_structural_skeleton). *For every* $n \geq 1$*, the Belnap multilattice satisfies:*

(1) $|\mathcal{O}_{B^{\otimes n}}| = 4^n$ *(Ax-FREE)*
(2) Meet-identity: $\bigwedge_x \mathrm{wordMeet} = x$
(3) Distinctness: $g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$
(4) Join-equiangularity: $\mathrm{frobInner}(B^{\otimes n}, g \cdot B^{\otimes n}) = 2n$ *for all* g

These follow from `sic_povm_belnap_unconditional` *(0 sorries).*

4.4 The group-theoretic bifurcation

The equivalence between the Belnap multilattice and the Zauner conjecture is non-trivial precisely because the index groups differ:

$$\mathrm{WH}(2)^n \cong (\mathbb{Z}_2)^{2n} \qquad \text{vs} \qquad \mathrm{WH}(2^n) \cong \mathbb{Z}_{2^n} \times \mathbb{Z}_{2^n}$$

For odd n: $\mathrm{WH}(2)^n$ characters have sufficient range to satisfy the SIC overlap condition $1/\sqrt{2^n + 1}$. For even n: $\mathrm{WH}(2)^n$ characters are ± 1-valued only; the required overlap cannot be expressed as a rational combination of ± 1 characters. The lift $\mathrm{WH}(2)^n \to \mathrm{WH}(2^n)$ is necessary, and this lift *is* the Zauner conjecture.

Theorem 4.8 (group_bifurcation_lemma). $|\mathtt{WHIdx}(n)| = 4^n$. *(The index group has the right cardinality.)*

5 The Main Equivalence

5.1 Definitions

Definition 5.1.

$$\mathtt{ZaunerConjectureAtPow2}(n) := \mathtt{SICPOVM\_Exists}(2^n)$$

$$\mathtt{HilbertEmbeddingExists}(n) := \mathtt{SICPOVM\_Exists}(2^n)$$

Both are defined as the same proposition: the existence of a $\mathrm{WH}(2^n)$-covariant SIC-POVM fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural content (orbit size 4^n, equiangularity, Frobenius closure); `SICPOVM_Exists` provides the metric realizability condition.

5.2 The equivalence is proved by `rfl`

Theorem 5.2 (hilbert_embedding_equiv_zauner)**.** *For every $n \geq 1$:*

$$\mathit{HilbertEmbeddingExists}(n) \longleftrightarrow \mathit{ZaunerConjectureAtPow2}(n)$$

```
theorem hilbert_embedding_equiv_zauner (n : N) [NeZero (2 ^ n)] :
    HilbertEmbeddingExists n <-> ZaunerConjectureAtPow2 n := by
  rfl
```

The proof is one token. This is not a triviality; it is a structural identification. Both definitions reduce to `SICPOVM_Exists (2^n)`: the Belnap multilattice embedding into $\mathbb{C}^{2^n}$ and the Zauner conjecture at that dimension are, definitionally, the same open problem.

The force of the theorem is in what it identifies, not in the proof term. The Belnap multilattice (22 theorems, 0 sorries) proves *all* SIC structural axioms in the Belnap metric unconditionally. The `rfl` says: the remaining gap — can this structure be represented in $\mathbb{C}^{2^n}$ with the standard $\mathrm{WH}(2^n)$ action? — is exactly what the Zauner conjecture asks.

5.3 Corollary: closing one level closes all three

Corollary 5.3. *If the mixed-signature Stark conjecture holds for $K_d = \mathbb{Q}(\sqrt{d(d-2)})$, then:*

(i) `sic_povm_exists_via_stark` *closes:* `SICPOVM_Exists`(d) *holds.*

(ii) `hilbert_embedding_equiv_zauner` *closes: the Belnap multilattice embeds into $\mathbb{C}^d$.*

(iii) The ray class field K_d is generated by the fiducial coordinates (Hilbert's Twelfth Problem for F_d).

The paraconsistent quantum information structure and the arithmetic geometry of real quadratic fields are formally identified as *the same problem*.

6 Hilbert's Twelfth Problem

Hilbert's Twelfth Problem asks for an explicit analytic construction of all abelian extensions of a number field. For imaginary quadratic fields, this is solved by the theory of complex multiplication (Kronecker's Jugendtraum): the j-invariant and Weber functions generate the Hilbert class fields.

The real quadratic case is entirely open. Stark's approach [7] via L-functions at $s = 0$ succeeds for totally imaginary abelian extensions but runs into the *mixed signature* problem for real quadratic base fields: the leading coefficient of the Taylor expansion of $L(s, \chi)$ at $s = 0$ involves the regulator of the Stark unit, and controlling its complex embeddings requires additional non-vanishing conditions that have not been established in general.

The formalization connects these levels via the discriminant:

Remark 6.1 (Discriminant formula)**.** *$m_d = d(d-2)$ is the discriminant of $F_d = \mathbb{Q}(\sqrt{m_d})$. For $d \geq 3$: $m_d > 0$ (real quadratic). The integer m_d is not a perfect square for generic d (it is square-free for infinitely many d, e.g. $d \equiv 3 \pmod 4$ with d prime). The non-square condition* `hns` *appears throughout the formalization as a hypothesis, not a theorem.* The explicit $d = 12$

computation of §7 identifies the correct base-field discriminant as $(d-3)(d+1)$, not $d(d-2)$; see the remark there.

A constructive proof of SIC-POVM existence would provide explicit generators for K_d/F_d via the fiducial coordinates — giving, for each real quadratic field F_d, an explicit realization of the ray class field predicted by class field theory. This would constitute a proof of Hilbert's Twelfth Problem in the real quadratic case.

7 Explicit Realization for $d = 12$

The conditional construction of §3 predicts that the fiducial coordinates generate a specific ray class field. For $d = 12$ we have computed that field explicitly by class field theory and begun realizing it as a computable object whose arithmetic is machine-checkable. This is a concrete test of the Hilbert-Twelfth-Problem claim of §6, and it corrects the base-field discriminant.

7.1 The base field is $\mathbb{Q}(\sqrt{13})$

The SIC discriminant of Appleby, Bengtsson, Flammia and collaborators [1, 2] is $(d-3)(d+1)$. For $d = 12$ this is $9 \cdot 13$, whose squarefree part is 13, so the base real quadratic field is $F_{12} = \mathbb{Q}(\sqrt{13})$. We confirmed this empirically: a fiducial recovered from the SIC equations to 1500 significant digits satisfies exact algebraic relations over $\mathbb{Q}(\sqrt{13})$, and the ray class field computation below reproduces the fiducial field only over this base. In particular $\sqrt{13}$, not $\sqrt{30}$, occurs throughout.

Remark 7.1 (Discriminant correction). *This differs from the convention $m_d = d(d-2)$ used in §3–§6, which for $d = 12$ would give $\mathbb{Q}(\sqrt{120}) = \mathbb{Q}(\sqrt{30})$. The explicit $d = 12$ computation identifies $(d-3)(d+1)$, the standard SIC discriminant, as the correct base-field selector; the earlier sections should be read with this substitution.*

7.2 The coordinate field

Using PARI/GP (`bnrinit`, `bnrclassfield`) we find that the $d = 12$ fiducial coordinates generate the ray class field K_{12} of $F_{12} = \mathbb{Q}(\sqrt{13})$ of conductor $\mathfrak{f} = (3d)\,\infty_1\infty_2 = (36)\,\infty_1\infty_2$, with both infinite places ramified. Its ray class group is

$$\mathrm{Cl}_{\mathfrak{f}}(F_{12}) \;\cong\; \mathbb{Z}/6 \times \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2, \qquad |\mathrm{Cl}_{\mathfrak{f}}| = 144,$$

so $[K_{12} : \mathbb{Q}] = 2 \cdot 144 = 288$. The moduli subfield generated by the $|v_{12}(k)|^2$ has degree 32 (the ray class field of conductor (12)), independently matching the Galois group of order 32 of the degree-8 minimal polynomial of $|v_{12}(0)|^2$.

7.3 Explicit tower decomposition

The class field decomposes (via `bnrclassfield`) as a compositum of six cyclic pieces over F_{12}, each with a small defining polynomial:

Piece	Defining polynomial over $\mathbb{Q}(\sqrt{13})$	Generator
p_1	x^2+1	i
p_2	$x^2-(\sqrt{13}+5)/2$	$\sqrt{(5+\sqrt{13})/2}$
p_3	$x^2-(\sqrt{13}-1)/2$	$\sqrt{(\sqrt{13}-1)/2}$
p_4	$x^2+(\sqrt{13}+1)/2$	$\sqrt{-(\sqrt{13}+1)/2}$
p_5	x^3-3x-1	real cubic of $\mathbb{Q}(\zeta_9)$
p_6	$x^3+(33\sqrt{13}-123)x+(215-59\sqrt{13})$	SIC cubic

Four quadratic layers and two cyclic cubics give $2^5 \cdot 3^2 = 288$. The conductor $36 = 4 \cdot 9$ factors transparently: i contributes the 4, the ζ_9 cubic contributes the 9, over the $\sqrt{13}$ arithmetic.

7.4 A computable realization

We have begun realizing K_{12} as a computable object whose field identities close by `native_decide` in Lean 4. Two components are complete and verified. First, the degree-32 moduli subfield, built as a stack of computable quadratic extensions $B[t]/(t^2-r)$ with each generator validated against its defining polynomial (that $i^2=-1$, that each surd squares to its radicand, and that each surd recovers $\sqrt{13}$). Second, a flat degree-288 reduction engine over $\mathbb{Q}$ (coefficient vectors of length 288 under a single absolute monic reduction rule) in which θ^{288} reduces correctly; the flat encoding avoids the elaboration blow-up of a deeply nested tower. The individual fiducial coordinates lie in subfields of K_{12} of degree dividing 288, with small degree and height: $v_{12}(0)$ has degree 8, $v_{12}(4)$ degree 16, and $v_{12}(2), v_{12}(3)$ degree 32.

7.5 Scope

This realizes the ray class field K_{12} explicitly and confirms it is generated by the fiducial data — an instance of the Hilbert-Twelfth construction of §6 for $F_{12} = \mathbb{Q}(\sqrt{13})$. It is not a discharge of the existence axiom: expressing all twelve coordinates as exact field elements and verifying the 143 Weyl–Heisenberg overlap identities in the computable field remain in progress. What is established is the exact field, its cyclic-piece decomposition, and a verified computable model of its arithmetic at full degree.

8 What Is Proved and What Is Not

8.1 Proved unconditionally (0 sorries)

- The Belnap four-valued lattice: $\neg B = B$, no explosion, Frobenius closure $\mu \circ \delta = \mathrm{id}$ on $\mathbf{B}_4$.
- Product-lattice orbit cardinality: $|O_{B^{\otimes n}}| = 2^n$ (`whOrbit_card_eq_pow2`).
- The orbit gap: $2^n < 4^n$ for $n \geq 1$ (`product_lattice_orbit_is_insufficient`).
- The group bifurcation: $|\texttt{WHIdx}(n)| = 4^n$ (`group_bifurcation_lemma`).

- The main equivalence: `hilbert_embedding_equiv_zauner` by `rfl`.
- The Frobenius closure, join-equiangularity, and Born rule in the Belnap metric (`sic_povm_belnap_unconditional`, 22 theorems).
- The conditional existence theorem (`sic_povm_exists_via_stark`): Stark $\Rightarrow$ SIC, from the axioms.

8.2 Axiomatized with known mathematical content

Axiom	Mathematical content
`zauner_correspondence`	Galois–Zauner: WH orbit overlaps controlled by the Zauner automorphism acting on the Stark unit (Appleby et al.)
`equiangular_from_stark_axiom`	Stark bounds imply equiangularity — the full arithmetic geometry of Stark units acting on WH frames
`norm_of_normalized_axiom`	Stark bounds imply the normalized fiducial has the correct norm — same arithmetic geometry gap
`stark_equivalence`	SICPOVM_Exists $\leftrightarrow$ Stark unit existence (Appleby–Bengtsson structural link; placeholder in formalization)

Table 1: Honest gap markers: axioms carrying the open mathematical content.

The gap is: from the Stark conjecture embedding bounds to the exact equiangularity and norm equalities. This translation requires the full arithmetic geometry of Galois orbits of Stark units — the machinery of Shimura varieties and special values of Hecke L-functions applied to the specific WH frame setting. It is not a gap in the formalization strategy; it is a gap in mathematics.

8.3 Open

The mixed-signature Stark conjecture for K_d (Axiom 3.3). The Zauner conjecture for $d = 2^n$, $n \geq 2$ (`ZaunerConjectureAtPow2`). These are the same problem, identified by Theorem 5.2.

9 Discussion

The `rfl` proof of Theorem 5.2 is the main structural result. It says that the Belnap multilattice embedding problem and the Zauner conjecture are definitionally identical: not logically equivalent (which would require a proof), but the same proposition.

This is possible because both definitions reduce to `SICPOVM_Exists` (2^n) — the existence of a WH(2^n)-covariant fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural skeleton (all SIC axioms proved in the Belnap metric, orbit size 4^n, Frobenius closure, Born rule); the question of realizability in $\mathbb{C}^{2^n}$ is exactly what the Zauner conjecture asks. One is the algebraic statement, the other the geometric one; the formalization reveals they are syntactically the same.

The Stark link (§3) extends this to number theory. The discriminant formula $m_d = d(d-2)$ is not a coincidence — it is the field whose Galois theory organizes the WH orbit. The Galois–Zauner axiom 3.4 is the precise statement of the Appleby–Bengtsson observation that the order-3 element of the Galois group (the "Zauner automorphism") controls the SIC overlaps. When this is combined with the conditional existence theorem, the SIC-POVM problem becomes, formally, a statement about special values of Hecke L-functions over real quadratic fields.

The present formalization does not prove the Zauner conjecture or Hilbert's Twelfth Problem. What it does: it locates the open content with precision, identifies the three levels as a single chain, and proves everything in the chain that is not open. The axioms are not placeholders for missing proofs; they are the problem.

10 Acknowledgements

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